

Sean Michael Manger

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SUMMARY

Electrical engineer with hands-on experience in FPGA design, RTL development, and hardware research across the defense industry and academic settings, seeking to apply this technical foundation to the defense sector/private industry.

EDUCATION

University of California
BS Electrical Engineering, Cum Laude

Santa Cruz, CA
June 2026

EXPERIENCE

Electrical Engineering Intern

June 2026 – Present

Naval Postgraduate School

Monterey, CA

- Conducted experiments and collected empirical data to reduce electromagnetic interference (EMI) and ensure compliance with MIL-STD requirements for Navy shipboard systems
- Designed and implemented EMI filtering solutions under senior engineer guidance, contributing to MILSPEC compliance efforts
- Configured and calibrated multi-channel oscilloscope setups to simultaneously capture data from multiple sources, improving data collection efficiency and productivity
- Refactored and optimized Verilog RTL code to support EMI experimentation

Undergraduate Researcher

July 2025 – Present

Hardware Systems Collective Lab

Santa Cruz, CA

- Implemented a novel fine-grain tuning method for a Time-to-Digital Converter (TDC) on FPGA, improving timing resolution as part of Pentimento research
- Ported and upgraded a legacy TDC design to the Zynq UltraScale+ platform, refactoring RTL/Verilog to meet new timing constraints and resource requirements
- Independently led the project from design through hardware validation, culminating in a written senior thesis

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Verilog, HTML, CSS

Tools: MATLAB, Simulink, Vivado, PSpice, Altium Designer

Libraries: Pynq, Pandas, Matplotlib, Seaborn, HDL Coder, Control Systems Toolbox

Languages: English (**Native**), Russian (**Elementary Proficiency**), Spanish (**Limited Working Proficiency**)

PROJECTS

Senior Thesis

January 2026 – May 2026

Time-to-Digital Converters

Verilog, Vivado, Python, Pynq, Pandas, Seaborn

- Developed a Python script to auto-generate PLL configurations based on Zynq UltraScale+ hardware constraints and user-specified input parameters
- Built a Python API using PYNQ to program DRP (Dynamic Reconfiguration Port) registers at a low level, reverse-engineering register behavior from limited/incomplete Xilinx documentation
- Created a semi-automated experimental workflow to collect hardware data traces, paired with custom plotting scripts for data visualization and analysis
- Achieved femtosecond resolution with the new sensor design, an improvement over the previous picosecond-resolution sensor

HOBBIES & INTERESTS

Technical Interests: FPGAs, Controls, Embedded Systems, Robotics, Aircraft, Unmanned Systems

Personal Hobbies: Reading, Language Learning, Cooking, Amateur Historian